

Digital IC Design Diploma

ASYC FIFO Assignment

Under supervision of:

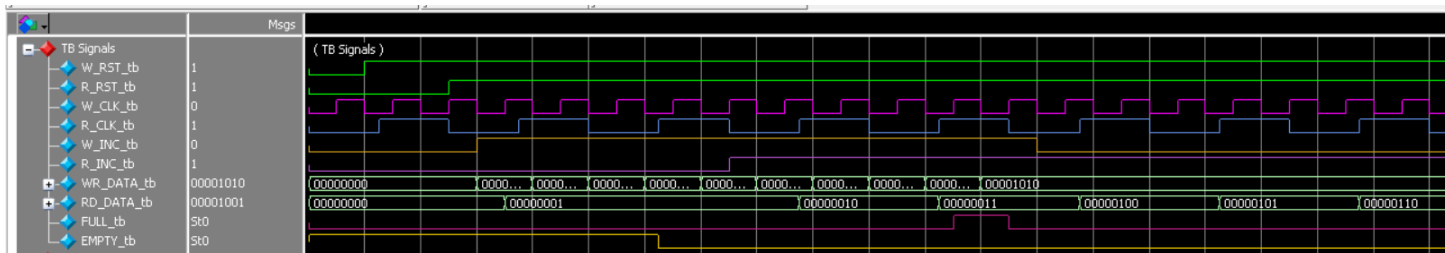
Eng. Ali Eltemsah

Submitted by:

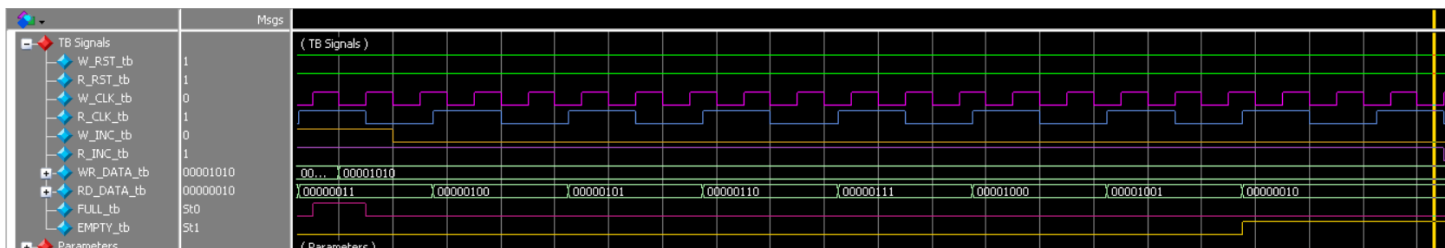
Youssef Ihab Mohamed

- **Test Cases Snippets:**

1) Writing data into FIFO then begin reading and observe full flag



2) Reading All the data and observe empty flag



3) Transcript Results

```
# -----
# Starting Simulation for ASYNC_FIFO
# Wrote 0x1 to FIFO mem in address 0
# Wrote 0x2 to FIFO mem in address 1
# Wrote 0x3 to FIFO mem in address 2
# Wrote 0x4 to FIFO mem in address 3
# Wrote 0x5 to FIFO mem in address 4
# Wrote 0x6 to FIFO mem in address 5
# Wrote 0x7 to FIFO mem in address 6
# Checking index 0. Read: 0x1, Expected: 0x1
# Test Case Passed
# Wrote 0x8 to FIFO mem in address 7
# Wrote 0x9 to FIFO mem in address 0
# Checking index 1. Read: 0x2, Expected: 0x2
# Test Case Passed
# Checking index 2. Read: 0x3, Expected: 0x3
# Test Case Passed
# Checking index 3. Read: 0x4, Expected: 0x4
# Test Case Passed
# Checking index 4. Read: 0x5, Expected: 0x5
# Test Case Passed
# Checking index 5. Read: 0x6, Expected: 0x6
# Test Case Passed
# Checking index 6. Read: 0x7, Expected: 0x7
# Test Case Passed
# Checking index 7. Read: 0x8, Expected: 0x8
# Test Case Passed
# Checking index 0. Read: 0x9, Expected: 0x9
# Test Case Passed
# Checking index 1. Read: 0x2, Expected: 0x2
# Test Case Passed
# -----
# Simulation Finished
# ** Note: $stop : ASYNC_FIFO_tb.v(91)
# Time: 474 ns Iteration: 0 Instance: /ASYNC_FIFO_tb
# Break in Module ASYNC_FIFO_tb at ASYNC_FIFO_tb.v line 91
```